

A new equilibrium temperature inflow profile for modelling the non-isothermal atmospheric boundary layer in CFD simulation

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ABSTRACT

The atmospheric boundary layer (ABL) flow in urban morphologies, ranging from micro-scale to local-scale, will become non-isothermal due to the combined effects of solar radiation, wall heat transfer, the urban heat island, and anthropogenic heat, resulting in a mutually coupled urban thermal and wind environment. An appropriate temperature inflow profile is the key prerequisite for Computational Fluid Dynamics (CFD) simulations of the non-isothermal ABL flow and the urban thermal and wind environment, as it must coincide with actual thermal characteristics while being compatible with CFD. Based on the equilibrium ABL model, a new equilibrium temperature inflow profile compatible with the energy equation, the standard gradient diffusion hypothesis, and the turbulence model is derived, which is proportional to the turbulent kinetic energy inflow profile, with the ratio further determined by the wall heat flux, demonstrating that turbulent thermal diffusion is strongly related to turbulent characteristics and atmospheric stability. To address the scaled effect, its full expression is provided based on thermodynamic similarity criteria. Using a numerical model of empty domain based on the SST $k-\omega$ model, the horizontal homogeneity of the equilibrium thermal and wind field modeled by the new set of inflow profiles is verified. Taking the flow around a single square building model as an example, its applicability in simulating the urban thermal and wind environment around buildings has been numerically validated, where the comparison of numerical and test results reveals tolerable errors overall.

1. Introduction

In urban regions with dense tall buildings, artificial underlying surfaces with distinct physical characteristics have extensively replaced the original natural surfaces like soil, vegetation, and water. This transformation results in various urban morphologies such as building blocks, street canyons, courtyards, and open spaces. Generally, the horizontal length scale of these urban morphologies ranges from 10 m to 1 km, spanning from micro-scale to local-scale in urban microclimate studies [1,2]. Under the combined effects of multiple factors such as solar radiation, wall heat transfer, the urban heat island, and anthropogenic heat, the Atmospheric Boundary Layer (ABL) flow at this scale will become non-isothermal (non-neutral). At this point, a complex interaction between aerodynamic and thermodynamic processes occurs, ultimately leading to the formation of the mutually coupled Urban Thermal and Wind Environment (UTWE).

The buoyancy effect of the UTWE will intensify the non-isothermal state and the instability of the ABL, directly impacting its characteristics such as the shape of the wind profile, the boundary layer thickness,

and wind veering. These changes, in turn, affect the wind fields around buildings and structural wind loads [3–5]. Moreover, the UTWE will impact the energy consumption of indoor cooling and mechanical ventilation, thereby increasing the energy costs and carbon emissions of buildings throughout its lifecycle [6–8]. Most importantly, the UTWE is a key component of the living environment, which is of great significance to public health and pedestrian comfort [9–12].

The accurate description of non-isothermal ABL flow and the corresponding UTWE is crucial for studies in wind engineering. In wind tunnel tests, the surfaces of ground or scaled models should be heated to specified temperatures to ensure thermodynamic similarity. Previously, through thermally stratified wind tunnel tests, Uehara et al. [13], using the Richardson number (Ri) to characterize the thermodynamic similarity, studied the impact of heated ground on wind environment in street canyons across 7 cases (Ri ranges from -0.21 to 0.79). Richards et al. [14] studied the impact of heated leeward surface of models on wind environment, then discussed energy consumption of indoor cooling based on test results. Allegrini et al. [15] and Lin et al. [16] employed the Particle Image Velocimetry technique to study the impact

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of non-uniform heating caused by solar radiation on the thermal and wind environment in street canyons.

Above test studies examined the thermal and wind environment issues using ideal urban morphological model under a weakly unstable state, where Ri is larger than -0.25 . However, in real urban morphologies, the non-isothermal ABL flow often reaches strongly unstable state, with Ri potentially smaller than -0.50 , especially during the summer. To ensure thermodynamic similarity under these conditions, it is necessary to heat the ground or scaled model surfaces into a very high temperature, like 400K, in wind tunnel tests, which is too difficult and dangerous to achieve. Therefore, scaled wind tunnel tests with real urban morphological models are rare.

To overcome this bottleneck of test, numerical simulation methods based on Computational Fluid Dynamics (CFD), including Reynolds-Averaged Navier-Stokes equations (RANS) and Large Eddy Simulation (LES), are becoming effective approaches for studying the thermal and wind environment. Their boundary condition settings are more flexible; thus, they can easily address the constraints of thermodynamic similarity. Jiang and Yoshie [17] simulated the thermal and wind environment of a building block model, validating that the numerical results agreed with the test results. Zhou et al. [4] simulated the non-isothermal flow around a single square building model across 6 cases, with the ground temperature modeled as 399.26K in the strongly unstable state case ($Ri = -1.50$).

In CFD simulations, providing an appropriate description of ABL characteristics at the inflow boundary is always a key challenge, particularly when describing temperature for simulating the UTWE under non-isothermal ABL flow. Previous ABL models commonly proposed inflow profiles describing mean wind speed U , turbulence kinetic energy k , and turbulent dissipation rate ε , which, strictly speaking, are only applicable to the isothermal (neutral) state [18–20]. Potsis et al. [21] emphasized the need to extend ABL models into the non-isothermal state to improve the accuracy of UTWE simulations. Accomplishing this requires an additional appropriate temperature inflow profile. Franke et al. [22] suggested using temperature profiles obtained from field measurements. Pontiggia et al. [23] and Pieterse and Harms [24] proposed mesoscale-appropriate temperature profiles based on meteorological theories and parameters. Many CFD simulation studies of the non-isothermal ABL flow and the UTWE have used uniform temperature profile at the inflow boundary [25–29]. Unfortunately, these temperature profiles are inadequate for accurate simulation because of their incompatibility with CFD.

In CFD simulations of the thermal and wind environment, it is essential that temperature inflow profile can not only reflect the actual vertical temperature distribution of ABL in urban morphologies accurately, but also be compatible with the energy equation, gradient diffusion hypothesis, current ABL model, wall function, aerodynamic roughness length, and more. Any incompatibility here could result in horizontal inhomogeneity of the temperature profile, leading to significant numerical errors [30]. Toparlak et al. [30] proposed a compatible temperature inflow profile based on RH model [18]. Recently, using this profile, a study of the thermal and wind environment has shown promising simulation results [11]. However, the uniform k inflow profile of the RH model clearly contradicts actual ABL flow. Furthermore, the RH model requires the imposition of artificial shear stress at the upper face of computational domain, which is an additional external intervention that complicates simulation and deviates from actual ABL flow again.

Hence, there is a need to develop a new compatible temperature inflow profile based on a more accurate ABL model to simulate non-isothermal ABL flow more accurately, ensuring horizontal homogeneity (sustainability), thereby enhancing the accuracy of CFD simulation for the UTWE under non-isothermal ABL flow and laying the groundwork for their evaluation and optimization.

In section 2, referencing the study of Toparlak et al. [30], a new equilibrium temperature inflow profile is theoretically and compatibly

derived based on the equilibrium ABL model proposed by authors. In section 3, using a numerical model of empty domain based on SST $k-\omega$ model, the new equilibrium temperature inflow profile is verified for horizontal homogeneity (sustainability) across entire computational domain. In section 4, taking the flow around a single square building model as an example, its applicability in simulating the coupled thermal and wind environment around buildings is validated through comparison of numerical and test results. Finally, section 5 provides a brief conclusion to this article.

Nomenclature

Roman symbols

c_p	Specific heat capacity (J/kg·K)
C_s	Roughness constant
C_μ	Turbulence model constant
C_1	Constant in equilibrium ABL model
C_2	Constant in equilibrium ABL model
F_1	Hyperbolic tangent blending function
g	Gravitational acceleration ($=9.8 \text{ m/s}^2$)
Gr	Grashof number
H	Building height (m)
k	Turbulent kinetic energy (m^2/s^2)
$k(z)$	Inflow profile of k (m^2/s^2)
k_{ref}	Reference turbulent kinetic energy (m^2/s^2)
K_s	Equivalent sand-grain roughness height (m)
l_s	Scaling factor of length scale
P_k	The production of k
Pr_t	Turbulent Prandtl number
q_{wall}	Wall heat flux (W/m^2)
R	Volumetric heat source (W/m^3)
Re	Reynolds number
Ri	Richardson number
Rib	Bulk Richardson number
t	Time (s)
$\mathbf{T}$	Temperature vector
T	Mean temperature (K)
$T(z)$	Inflow profile of T (K)
T'	Fluctuating temperature (K)
T_g	Ground temperature (K)
$T_{g, m}$	Ground temperature of scaled model (K)
$T_{g, p}$	Ground temperature of prototype (K)
$T_m(z)$	Inflow profile of T for scaled model (K)
T_{ref}	Reference temperature (K)
U	Mean wind speed (m/s)
$U(z)$	Mean wind speed inflow profiles (m/s)
U_{ref}	Reference wind speed (m/s)
$\bar{u}$	Mean streamwise velocity (m/s)
u'	Fluctuating streamwise velocity (m/s)
u_*	ABL friction velocity (m/s)
u_H	Reference $\bar{u}$ at H (m/s)
$\mathbf{V}$	Velocity vector
$\bar{v}$	Mean spanwise velocity (m/s)
$\bar{w}$	Mean vertical velocity (m/s)
w'	Fluctuating vertical velocity (m/s)
x	Streamwise axis coordinate (m)
y	Spanwise axis coordinate (m)
z	Height above the ground surface (m)
z_0	Aerodynamic roughness length (m)
z_{ref}	Reference height (m)
z^+	Normalized height

Greek symbols

α	Thermal diffusivity (m^2/s)
α_t	Turbulent thermal diffusivity (m^2/s)
β	Constant in SST $k-\omega$ model
β_1	Constant in Wilcox $k-\omega$ model
β_2	Constant in standard $k-\varepsilon$ model
γ	Constant in SST $k-\omega$ model
γ_1	Constant in Wilcox $k-\omega$ model
γ_2	Constant in standard $k-\varepsilon$ model
ε	Turbulent dissipation rate (m^2/s^3)
$\varepsilon(z)$	Inflow profile of ε (m^2/s^3)
θ	Potential temperature (K)
Θ	Normalized temperature
$\Theta(z)$	Normalized temperature profile
Θ_m	Θ profile for scaled model
Θ_p	Θ profile for prototype
κ	Von Karman constant ($=0.41$)

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λ	Thermal conductivity (W/m·K)
$\lambda_{(T_{ref}-T_g)}$	Scaled factor of $(T_{ref}-T_g)$
λ_U	Scaled factor of U
μ_t	Eddy viscosity (kg/m·s)
ρ	Density of air (kg/m ³)
σ_k	Prandtl number of k in SST $k-\omega$ model
σ_{k1}	Prandtl number of k in Wilcox $k-\omega$ model
σ_{k2}	Prandtl number of k in standard $k-\varepsilon$ model
σ_ω	Prandtl number of ε in SST $k-\omega$ model
$\sigma_{\omega1}$	Prandtl number of ε in Wilcox $k-\omega$ model
$\sigma_{\omega2}$	Prandtl number of ε in standard $k-\varepsilon$ model
Φ	Constants set of SST $k-\omega$ model
Φ_1	Constants set of Wilcox $k-\omega$ model
Φ_2	Constants set of standard $k-\varepsilon$ model
ω	Specific dissipation rate (1/s)
$\omega(z)$	Inflow profile of ω (1/s)
Acronyms	
ABL	Atmospheric Boundary Layer
AIJ	Architectural Institute of Japan
CFD	Computational Fluid Dynamics
LES	Large Eddy Simulations
RANS	Reynolds-Averaged Navier-Stokes
TPU	Tokyo Polytechnic University
SGDH	Standard Gradient Diffusion Hypothesis
UTWE	Urban Thermal and Wind Environment

2. Derivation of new equilibrium temperature inflow profile

2.1. The equilibrium ABL model

An accurate and compatible model of the ABL at inflow boundary ensuring horizontal homogeneity is the key prerequisite for CFD simulations of wind load and environment. Under the isothermal (neutral) state, the equilibrium ABL model was proposed by the authors [18,19] based on RANS, with the mathematical form as follows:

$$U(z) = \frac{u_*}{\kappa} \ln \left(\frac{z + z_0 l_s}{z_0 l_s} \right) \quad (1)$$

$$k(z) = \frac{u_*^2}{\sqrt{C_\mu}} \sqrt{C_1 \ln \left(\frac{z + z_0 l_s}{z_0 l_s} \right) + C_2} \quad (2)$$

$$\varepsilon(z) = \frac{u_*^3}{\kappa(z + z_0)} \sqrt{C_1 \ln \left(\frac{z + z_0 l_s}{z_0 l_s} \right) + C_2} \quad (3)$$

$$\omega(z) = \frac{u_*}{\kappa \sqrt{C_\mu}} \frac{1}{z + z_0 l_s} \quad (4)$$

Where, $U(z)$, $k(z)$, $\varepsilon(z)$, and $\omega(z)$ denote the inflow profiles of mean wind speed U , turbulent kinetic energy k , turbulent dissipation rate ε , and specific dissipation rate ω , respectively; u_* the ABL friction velocity, κ ($=0.41$) the von Karman constant, z_0 the aerodynamic roughness length, C_μ ($=0.028$) constant in turbulence model, l_s the scaled factor defined as the ratio between the length scale of model and prototype, C_1 and C_2 constants in the equilibrium ABL model which can be determined by fitting wind tunnel test results or field measurement data.

The equilibrium ABL model coincides with the turbulent characteristics of actual ABL flow and is compatible with both the $k-\varepsilon$ turbulence model [31] and the $k-\omega$ turbulence model [32,33]. Studies have verified that the equilibrium ABL model can accurately model the isothermal ABL flow at the inflow boundary without extra modifications, with the streamwise gradients of $U(z)$, $k(z)$, $\varepsilon(z)$, and $\omega(z)$ sustaining almost zero throughout the entire domain, ensuring horizontal homogeneity [18, 19]. Recently, it was included in the famous open CFD code OpenFOAM [34] as one of the recommended inflow profiles.

2.2. New equilibrium temperature inflow profile

This subsection theoretically derives a new equilibrium temperature inflow profile based on the equilibrium ABL model to extend it into the non-isothermal (non-neutral) state. The new equilibrium temperature inflow profile is compatible across multiple aspects of CFD while ensuring the constraint of horizontal homogeneity.

2.2.1. Transformation of energy equation

In CFD simulation, the general form of energy equation is as follows:

$$\frac{\partial T}{\partial t} = \nabla \cdot (\alpha \nabla T) - \nabla \cdot (\mathbf{V}T) + R \quad (5)$$

Where, α denotes the thermal diffusivity, T the temperature vector, $\mathbf{V}$ the velocity vector, R the volumetric heat source. Assuming no extra anthropogenic heat, $R = 0$.

Based on the steady condition of RANS which requires $\partial T / \partial t = 0$, Eq. (5) can be simplified into the steady form as follows:

$$0 = \nabla \cdot (\alpha \nabla T) - \nabla \cdot (\mathbf{V}T) \quad (6)$$

Assuming no wind veering, the thermal and wind field of the ABL flow can be considered inhomogeneous only in the vertical direction, implying that the three-dimensional flow can be simplified to two-dimensional flow containing only the streamwise and vertical directions. Also, in this scenario, only in streamwise direction (x -direction), the mean velocity would have value, denoted as $\bar{u} \neq 0$, while $\bar{w} = 0$. Thus, combined with the principle of RANS, Eq. (6) can be simplified and expanded into the RANS form as follows:

$$\alpha \left(\bar{u} \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(\bar{u} \frac{\partial T}{\partial x} + \frac{\partial \bar{u} T}{\partial x} + \frac{\partial \bar{w} T}{\partial z} \right) = 0 \quad (7)$$

Where, T denotes the mean temperature, $\bar{u}$ the mean streamwise velocity.

Similar to the Reynolds stress, non-closure terms emerge in Eq. (7) after Reynolds averaging, namely the turbulent heat flux $(\bar{u}'T', \bar{w}'T')$. In the turbulence model, to close the momentum equation, the Eddy Viscosity Hypothesis is employed so that the Reynolds stress can be modeled by the eddy viscosity μ_t . Here, to close the term of turbulent heat flux in Eq. (7), the Standard Gradient Diffusion Hypothesis (SGDH) is employed, which assumes that the turbulent heat flux is proportional to the vertical gradient of mean temperature T , with the ratio defined as the turbulent thermal diffusivity α_t . Additionally, α_t is assumed to be isotropic and proportional to μ_t , with the ratio defined as the turbulent Prandtl number Pr_t . The mathematical expressions for the SGDH are as follows:

$$\bar{u}'T' = -\alpha_t \frac{\partial T}{\partial x}, \bar{w}'T' = -\alpha_t \frac{\partial T}{\partial z}, \alpha_t = \frac{\mu_t}{\rho Pr_t} \quad (8)$$

Under the SGDH, Pr_t represents the ratio of the turbulent contribution to momentum diffusivity and thermal diffusivity. For air, Pr_t ranges from about 0.70 to 0.90, and commonly takes 0.85 as the default value in CFD codes such as OpenFOAM [34] and ANSYS Fluent [35].

Substituting Eq. (8) into Eq. (7), while employing the constraint of horizontal homogeneity which requires the streamwise gradient of temperature $\partial T / \partial x$ equals to zero, Eq. (7) can be finally transformed as follows:

$$\frac{\partial}{\partial z} \left(\alpha \frac{\partial T}{\partial z} + \alpha_t \frac{\partial T}{\partial z} \right) = 0 \quad (9)$$

Which means:

$$\left(\alpha \frac{\partial T}{\partial z} + \alpha_t \frac{\partial T}{\partial z} \right) = \text{const} \quad (10)$$

As can be seen in Eq. (10), thermal diffusion in the non-isothermal

ABL flow arises from both natural mechanisms (represented by α) and turbulence (represented by α_t). The sum of their products with the vertical gradient of mean temperature remains constant.

2.2.2. Representation of the vertical gradient of mean temperature

Based on Fourier's Law, the vertical gradient of mean temperature under one-dimensional natural heat conduction is proportional to the wall heat flux per unit time and unit area, gives:

$$\alpha \frac{\partial T}{\partial z} = \frac{\lambda}{\rho c_p} \frac{\partial T}{\partial z} = -\frac{q_{\text{wall}}}{\rho c_p} \quad (11)$$

Where, λ denotes the thermal conductivity, q_{wall} the wall heat flux, c_p the specific heat capacity (about 1005 J/kg·K for air).

As shown in Eq. (11), α follows Fourier's law, which implies that $\alpha(\partial T / \partial z)$ equals a constant. Then combining Eq. (10) and Eq. (11) clearly indicates that $\alpha_t(\partial T / \partial z)$ must also be a constant. Since they are both constants, it is natural to assume that the behavior of α_t in thermal diffusion is similar to that of α . Thus, assuming that $\alpha_t(\partial T / \partial z)$ also follows Fourier's Law, which yields:

$$\alpha_t \frac{\partial T}{\partial z} = -\frac{q_{\text{wall}}}{\rho c_p} \quad (12)$$

Essentially, Eq. (12) means that, in the steady state, a hypothetical equilibrium between α_t and the wall heat flux q_{wall} is assumed, which was also characterized by the vertical gradient of mean temperature. Combining with the SGDH, substituting Eq. (8) into Eq. (12) yields:

$$\frac{\partial T}{\partial z} = -\frac{q_{\text{wall}} \text{Pr}_t}{c_p \mu_t} \quad (13)$$

As can be seen in Eq. (13), based on the Fourier's Law and the SGDH, the vertical gradient of T can be modeled by the eddy viscosity μ_t of turbulence model.

2.2.3. Derivation of new equilibrium temperature inflow profile

The equilibrium ABL model proposed by the authors is compatible with both the k - ε turbulence model [31] and the k - ω turbulence model [32,33], of which μ_t can be computed as follows [19,20]:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} = \rho \frac{k}{\omega} \quad (14)$$

Substituting the equilibrium ABL model (Eqs. (1)–(4)) into Eq. (14) yields μ_t as follows:

$$\mu_t = \rho u_* \kappa (z + z_0) \cdot \sqrt{C_1 \ln \left(\frac{z + z_0}{z_0} \right) + C_2} \quad (15)$$

Substituting Eq. (15) into Eq. (13) yields the vertical gradient of T , which is compatible with the equilibrium ABL model, as follows:

$$\frac{\partial T}{\partial z} = -\frac{q_{\text{wall}} \text{Pr}_t}{c_p \rho u_* \kappa} \cdot \frac{1}{(z + z_0) \cdot \sqrt{C_1 \ln \left(\frac{z + z_0}{z_0} \right) + C_2}} \quad (16)$$

By performing the indefinite integral of Eq. (16) along the vertical direction involving specific substitution, a new equilibrium temperature inflow profile with an undetermined constant term can be derived as follows:

$$T(z) = \int \frac{\partial T}{\partial z} (z + z_0) d(\ln(z + z_0)) = -\frac{q_{\text{wall}} \text{Pr}_t}{c_p \rho u_* \kappa} \cdot \frac{2}{C_1} \cdot \sqrt{C_1 \ln \left(\frac{z + z_0}{z_0} \right) + C_2} + \text{const} \quad (17)$$

Eq. (17) should equal the reference temperature T_{ref} at the reference height z_{ref} , thereby its undetermined constant term can be obtained as follows:

$$\text{const} = T_{\text{ref}} + \frac{q_{\text{wall}} \text{Pr}_t}{c_p \rho u_* \kappa} \cdot \frac{2}{C_1} \cdot \sqrt{C_1 \ln \left(\frac{z_{\text{ref}} + z_0}{z_0} \right) + C_2} \quad (18)$$

Noting that the temperature inflow profile shown in Eq. (17) and the turbulent kinetic energy inflow profile shown in Eq. (3) have the same term, which is the root square of $C_1 \ln(z + z_0 / z_0) + C_2$. Thus, combining and rearranging Eq. (3), Eq. (17) and Eq. (18), the new equilibrium temperature inflow profile can finally be derived as follows:

$$T(z) = T_{\text{ref}} - \frac{q_{\text{wall}} \text{Pr}_t}{c_p \rho u_* \kappa} \cdot \frac{2\sqrt{C_\mu}}{C_1 u_*^2} [k(z) - k_{\text{ref}}] \quad (19)$$

Where, k_{ref} denotes the reference turbulent kinetic energy at the reference height z_{ref} .

As can be seen in Eq. (19), the new equilibrium temperature inflow profile is proportional to the turbulent kinetic energy inflow profile, with the ratio further determined by the wall heat flux. This demonstrates that turbulent thermal diffusion is strongly related to turbulent characteristics and atmospheric stability.

During the process of theoretical derivation outlined above, the new equilibrium temperature inflow profile shown in Eq. (19) is theoretically compatible with the energy equation, the SGDH, the turbulence model, and the equilibrium ABL model one by one, while also theoretically ensuring the constraint of horizontal homogeneity. Throughout the entire derivation process, no extra modifications were made to the original inflow profiles of U , k , ε , and ω in the equilibrium ABL model (Eqs. (1)–(4)). That means, only by appropriately determining the reference temperature T_{ref} at the reference height z_{ref} and the wall heat flux q_{wall} , the new equilibrium temperature inflow profile can be uniquely determined, where there is no need of extra model constants.

2.3. Scaled effect

Wind tunnel tests for the non-isothermal ABL flow and the UTWE are typically conducted using scaled models, where the scaling factor l_s can be up to 1:100 or 1:300. In such cases, to ensure similarity of aerodynamic and thermodynamic between the scaled model and the prototype, it is essential to properly account for the scaled effect in the CFD simulation corresponding to the scaled wind tunnel test.

Under the isothermal (neutral) state, taking proper consideration of the scaled effect, the full expression of U , k , ε , and ω in the equilibrium ABL model have been provided by previous study of the authors [20], ensuring aerodynamic similarity. Similarly, for modelling non-isothermal ABL flow, the temperature inflow profile should also appropriately account for the scaled effect. Based on thermodynamic similarity criteria, this subsection extends the new equilibrium temperature inflow profile from its full-scale form (Eq. (19)) into its full expression addressing scaled effect.

Firstly, normalize the energy equation. Normalize the height and the temperature using the following equations to obtain the normalized height and the normalized temperature:

$$z^+ = z / z_{\text{ref}} \quad (20)$$

$$\Theta = \frac{T(z) - T_g}{T_{\text{ref}} - T_g} \quad (21)$$

Where, z^+ denotes the normalized height, Θ the normalized temperature. Eq. (21) is a common approach in the studies of the non-isothermal ABL flow and the UTWE, where Θ varies from 0 to 1 between the ground level and the reference height.

Substituting Eqs. (20) and (21) into the full-scale energy equation shown in Eq. (9) yields:

$$\frac{\partial \Theta}{\partial z^+} \left(\alpha \frac{\partial \Theta}{\partial z^+} + \alpha_t \frac{\partial \Theta}{\partial z^+} \right) = 0 \quad (22)$$

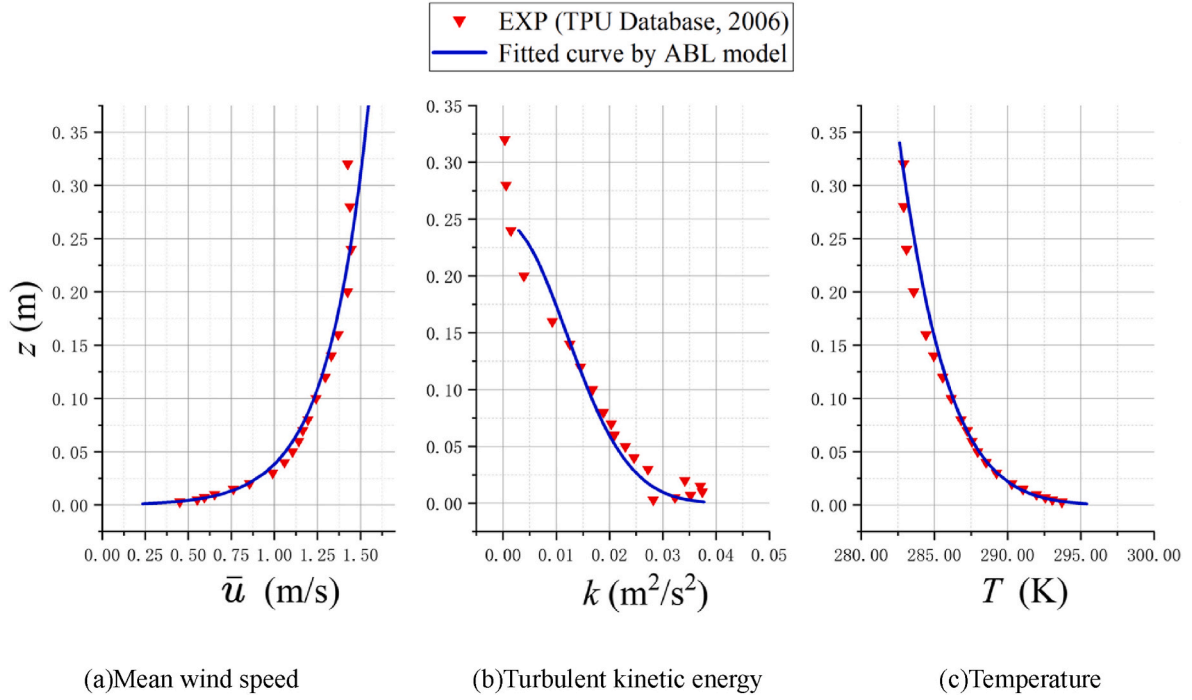


Fig. 1. Non-isothermal ABL inflow profile of wind tunnel test and corresponding fitted curves. (a) Mean wind speed (b) Turbulent kinetic energy (c) Temperature.

As can be seen in Eq. (22), the normalized energy equation is the same as Eq. (9) in terms of their form. That means, the normalized equilibrium temperature inflow profile can be theoretically derived using the same approach outlined in subsection 2.2, which yields:

$$\Theta(z) = 1 - \frac{q_{\text{wall}} \text{Pr}_t}{c_p \rho u_* \kappa} \cdot \frac{2\sqrt{C_\mu}}{C_1 u_*^2} \cdot \frac{1}{(T_{\text{ref}} - T_g)} [k(z) - k_{\text{ref}}] \quad (23)$$

Note that Eq. (23) can also be derived by normalizing its full-scale form (Eq. (19)) based on Eq. (21). For similarity, the normalized temperature Θ should be identical at the same normalized height z^+ . Following this criterion, the normalized temperature profile of the scaled model should be the same as that of the prototype. Hence, the full expression of the equilibrium temperature inflow profile to account for the scaled effect is as follows:

$$\Theta_m = \Theta_p \Leftrightarrow T_m(z) = (T_{\text{ref}} - T_{g,m})\Theta_p + T_{g,m} \quad (24)$$

Where, Θ_m denotes the normalized temperature inflow profile of scaled models, Θ_p that of prototype, $T_m(z)$ the equilibrium temperature inflow profile for scaled models, $T_{g,m}$ the scaled temperature of ground surface.

Obviously, Eq. (24) also requires determining $T_{g,m}$ to account for the scaled effect, which should be addressed by employing an additional dimensionless number with a corresponding similarity criterion. Since the thermal environment primarily affects the flow through the buoyancy effect, the similarity of the non-isothermal ABL flow and the UTWE can additionally be characterized by the Richardson number Ri .

The Richardson number Ri represents the ratio of the buoyancy term to the shear term of the flow, reflecting the instability between buoyancy and gravitation within the ABL by combining thermodynamic parameters such as atmospheric stability with aerodynamic parameters such as wind shear. In engineering studies, two commonly used variants of Ri are the gradient Richardson number and the bulk Richardson number (Rib). The expression for the gradient Richardson number is given as [36]:

$$Ri = \frac{g}{T} \frac{\partial \theta / \partial z}{(\partial \bar{u} / \partial z)^2 + (\partial \bar{v} / \partial z)^2} \quad (25)$$

Where, g denotes the gravitational acceleration, θ the potential temperature. Generally, it is recognized that $Ri = 0$ indicates a neutral state of atmospheric static stability, while $Ri = 0.25$ indicates a neutral state of atmospheric dynamic stability. In other words, when $Ri < 0$, the atmosphere is unstable; when $0 < Ri < 0.25$, the atmosphere is statically stable but dynamically unstable; and when $Ri > 0.25$, the atmosphere is stable.

The bulk Richardson number makes a finite difference approximation to the gradient Richardson number, where the partial differential term in Eq. (25) is replaced by the finite difference of variables, resulting in a more computationally convenient expression as follows [36,37]:

$$Rib = \frac{Gr}{Re^2} = \left(\frac{g}{T} \frac{\Delta T}{\Delta z} \right) / \left(\frac{(\Delta \bar{u})^2 + (\Delta \bar{v})^2}{\Delta z} \right) \quad (26)$$

Where, Gr denotes the Grashof number, Re the Reynolds number.

In studies of the UTWE, the reference height z_{ref} is generally taken as the building height H . Since we have already assumed that the thermal and wind field of the ABL flow can be considered inhomogeneous only in the vertical direction, $\bar{v} = 0$, and ΔT and $\Delta \bar{u}$ can be characterized using the differences in T and U within H (i.e., between the air at the roof and ground level), which yields:

$$Rib = \left(\frac{g}{T_{\text{ref}}} \frac{T_{\text{ref}} - T_g}{H} \right) / \left(\frac{U_{\text{ref}}}{H} \right)^2 \quad (27)$$

Where, T_{ref} denotes the reference temperature, T_g the ground temperature, U_{ref} the reference mean wind speed. This form of Rib is widely used in the studies of the non-isothermal ABL flow and the UTWE [4,13,38–40]. Due to such approximation, the neutral state of atmospheric stability can no longer be characterized by a unique value of Rib . Previous study suggested that the atmosphere is stable only when Rib is within the range of 0.5–1.0 [41].

Taking the scaled effect into account, the additional similarity criterion provided by Rib shown in Eq. (27) is:

$$\left(l_s \cdot \frac{\lambda_{(T_{\text{ref}} - T_g)}}{\lambda_U^2} \right) = 1 \quad (28)$$

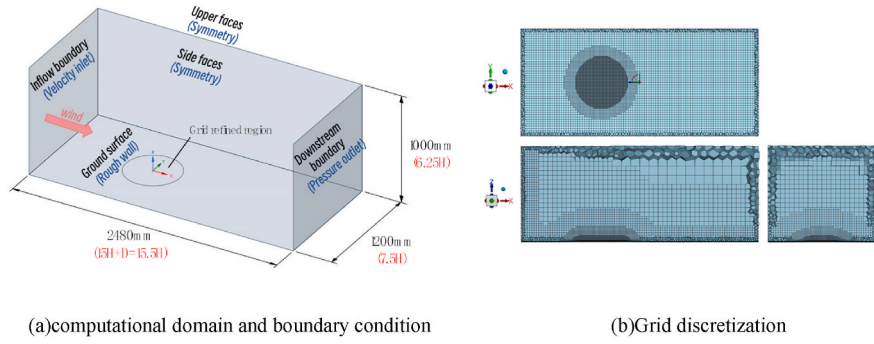


Fig. 2. Schematic diagram and information about Case A (a)computational domain and boundary condition (b)Grid discretization.

Table 1

Boundary conditions of numerical calculation.

Location	Boundary condition	Mathematical expression
Inflow boundary	Velocity inlet	Eqs. (31)–(35)
Downstream boundary	Pressure outlet	$\frac{\partial}{\partial x}(u, v, w, k, \omega, T) = 0$
Upper face of computational domain	Symmetry	$w = 0; \frac{\partial}{\partial x}(u, v, k, \omega, T) = 0$
Side faces of computational domain	Symmetry	$v = 0; \frac{\partial}{\partial x}(u, w, k, \omega, T) = 0$
Ground surface	Nonslip wall with roughness modification	$u = v = w = 0; K_s = 0.0035m; C_s = 0.75$

Where, $\lambda_{(T_{ref}-T_g)}$ denotes the scaled factor of the temperature difference between reference height and ground level, λ_U the scaled factor of the reference mean wind speed. Eq. (28) yields the $T_{g,m}$ can be obtained as follows:

$$T_{g,m} = T_{ref} - \frac{\lambda_U^2}{l_s} (T_{ref} - T_{g,p}) \quad (29)$$

In Eq. (29), l_s and λ_U are the same as typical scaled wind tunnel tests and CFD simulations under the isothermal (neutral) state. Substituting Eq. (29) into Eq. (24) yields:

$$T_m(z) = \frac{\lambda_U^2}{l_s} (T_{ref} - T_{g,p}) (\Theta_p - 1) + T_{ref} \quad (30)$$

Finally, as shown in Eq. (30), the full expression of the new equilibrium temperature inflow profile addressing scaled effect is theoretically derived based on two thermodynamic similarity criteria.

Here in Eqs. (24) and (30), T_{ref} takes the value of the modeled temperature at the reference height z_{ref} when conducting scaled wind tunnel tests and corresponding CFD simulations. Clearly, this modeled temperature may differ from the actual temperature. However, by adjusting $T_{g,m}$, it is undoubted that an identical Rib can still be achieved to ensure thermodynamic similarity. Essentially, Eq. (28) shows that, by appropriately determining the difference between T_{ref} and $T_{g,m}$, rather than their specific values separately, the thermodynamic similarity characterized by Rib can be maintained. And as shown in Eq. (29), if the modeled temperature matches the actual temperature, $T_{g,m}$ would have a corresponding value; otherwise, $T_{g,m}$ would take another value. Proper calculation of $T_{g,m}$ ensures effective management of the scaled effect on the modeled temperature compared to the actual temperature simultaneously.

3. Verification of horizontal homogeneity

In this section, with the conjunction of the equilibrium ABL model shown in Eqs. (1)–(4), the performance of the new equilibrium temperature inflow profile shown in Eq. (19) is verified by reproducing the non-isothermal ABL flow observed in a wind tunnel test conducted in the

thermally stratified wind tunnel at Tokyo Polytechnic University [42], ensuring the horizontal homogeneity of equilibrium thermal and wind field.

3.1. Fitted inflow profiles of equilibrium thermal and wind field

The test section of the thermally stratified wind tunnel at TPU is 10 m long, 1.2 m wide, and 1.0 m high. The ground surface consists of six heating and cooling panels, which can adjust temperatures to generate specified non-isothermal ABL inflow profiles of mean wind speed, turbulent kinetic energy, and temperature base on TPU test data and their corresponding fitted curves. During the TPU test [42], the ground temperature T_g was 318.45K; while at a reference height z_{ref} of 0.16 m, reference mean wind speed u_{ref} was 1.37 m/s, the reference temperature T_{ref} was 284.45K, the Reynolds number Re and the Rib was approximately 1.5×10^4 and -0.10 . The inflow profile of mean wind speed exhibited an exponential power of about 0.22 in the wind tunnel test.

As shown in the blue lines of Fig. 1, combining with the equilibrium ABL model (Eqs. (1)–(4)) and the new equilibrium temperature inflow profile (Eq. (19)), fitted curves of mean wind speed, turbulent kinetic energy, and temperature were obtained. The parameters of the fitted curves are as follows:

- (1) As shown in Fig. 1(a), for the mean wind speed curve fitted with the log law (Eq. (1)), $u_* = 0.10$ m/s, $\kappa = 0.41$ and $z_0 = 6 \times 10^{-4}$ m;
- (2) As shown in Fig. 1(b), for the turbulent kinetic energy curve fitted with Eq. (2), $C_1 = -0.079$ and $C_2 = 0.476$;
- (3) As shown in Fig. 1(c), for the temperature curve fitted with Eq. (19), $T_{ref} = 282.89K$, $k_{ref} = 0.041$ m²/s², and $q_{wall} = 136.5$ W/m² at the height of 0.32 m, while $C_1 = -0.19$ and $C_2 = 1.66$.

Substituting above fitted parameters into Eqs. (1)–(4) and Eq. (19) and taking $C_\mu = 0.028$ [19] yields the equilibrium expression of the non-isothermal ABL inflow profiles corresponding to the wind tunnel test shown in Fig. 1, as follows:

$$U(z) = 0.24 \ln \left(\frac{z + 0.0006}{0.0006} \right) \quad (31)$$

$$k(z) = \frac{0.01}{\sqrt{0.028}} \sqrt{-0.075 \ln \left(\frac{z + 0.0006}{0.0006} \right) + 0.476} \quad (32)$$

$$\varepsilon(z) = \frac{0.001}{0.41(z + 0.0006)} \sqrt{-0.075 \ln \left(\frac{z + z_0 l_s}{z_0 l_s} \right) + 0.476} \quad (33)$$

$$\omega(z) = \frac{0.10}{0.41 \sqrt{0.028} z + 0.0006} \quad (34)$$

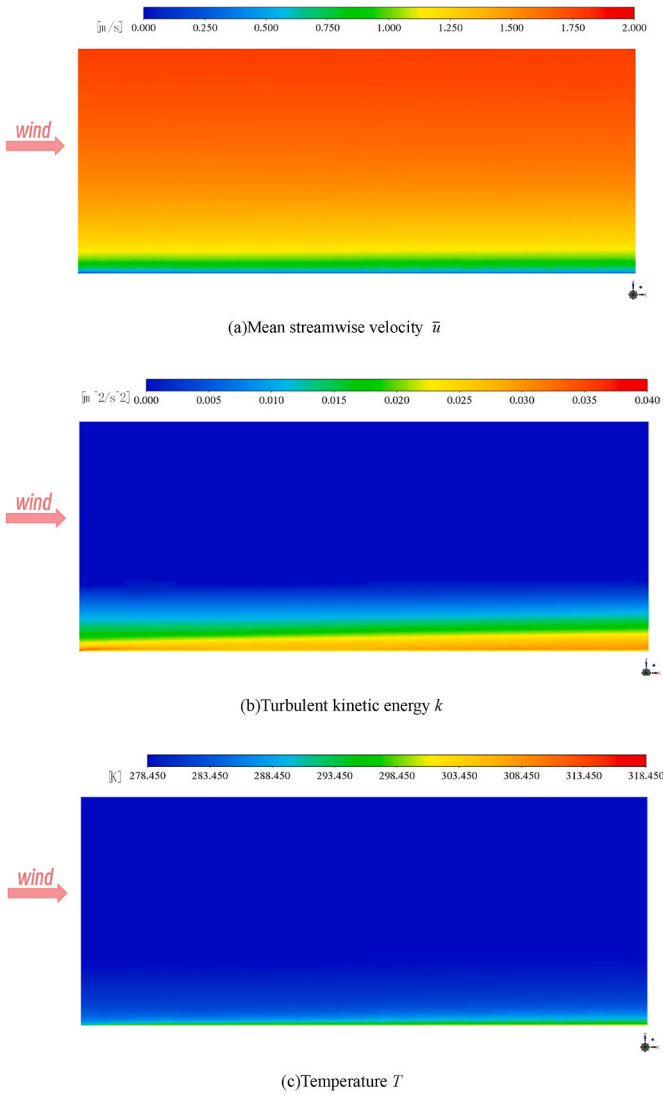


Fig. 3. Contours of Non-isothermal ABL flow on longitudinal plane ($y/H = 0$) in case A. (a) Mean streamwise velocity $\bar{u}$ (b) Turbulent kinetic energy k (c) Temperature T .

$$T(z) = 282.89 + 396.78[k(z) - 0.041] \quad (35)$$

where $C_1 = -0.19$; $C_2 = 1.66$

3.2. Numerical model of empty domain

According to present inflow profiles (Eqs. (31)–(35)), a 3D numerical simulation in a domain without obstacles was conducted, denoted as Case A. The computational domain for Case A, as shown in Fig. 2(a), has the length of 2.48 m, and the cross-sectional dimensions of 1.2 m width and 1.0 m height, which coincides with the thermally stratified wind tunnel. Table 1 lists the boundary conditions for Case A, as also shown in Fig. 2(a), where the inflow boundary is modeled as velocity inlet, the downstream boundary as pressure outlet, and the upper and side faces of computational domain as symmetry. The ground surface is modeled as rough non-slip wall with enhanced wall function. The two parameters to model rough wall, i.e., the equivalent sand-grain roughness height K_s and roughness constant C_s , are determined as $K_s = 0.0035$ m and $C_s = 0.75$ by referencing their relationship with z_0 derived by Blocken et al. [43].

Fig. 2(b) illustrates the 3D sectional view of the grid discretization for Case A. Polyhedral surface mesh is employed to discretize each face

of the computational domain, followed by filling the volume with hexahedral cells, maintaining a constant stretching ratio of 1.10. The maximum surface mesh sizes are set as follows: 48 mm for the inflow boundary, 12 mm for the ground surface, and 96 mm for the downstream boundary, upper face, and side faces.

After discretizing the polyhedral surface mesh, it transitions to hexahedral cells to fill the volume, with a maximum cell size of 96 mm. Additionally, grid refinement is applied around a hypothetical model location (near the origin), with hexahedral cells refined to a minimum size of 3 mm. Starting from the hypothetical model location (near the origin), grid cells grow outward radially layer by layer to fill the volume. The algorithm operates as follows: starting from the minimum refinement size (i.e., 3 mm), the cell size remains unchanged for several transition layers until the “nominal” size, calculated with the stretching ratio of 1.10 layer by layer, exceeds twice the current size level. At that point, the cell size increases to next level using a ‘two-in-one’ approach (combining 2 smaller cells into 1 larger cell). Repeat this procedure sequentially with the size levels of 3 mm, 6 mm, 12 mm, 24 mm, 48 mm, and 96 mm until the grid cells are fully discretized. The stretching ratio remains constant at 1.10. For each layer of cells, the nominal size accumulates until it exceeds twice the current size level.

Given that $1.1^7 < 2 < 1.1^8$, there are 8 transition layers where the cell size remains unchanged before transitioning to the next size level. Similarly, 8 layers of boundary layer elements are adopted above the ground surface, where the height of the first layer is set as 0.82 mm, and the vertical stretching ratio is also 1.10. Overall, the total amount of cells of Case A is about 600,000, with all cells exhibiting orthogonality quality greater than 0.50.

3.3. Numerical method and turbulence model constants

Steady numerical simulation of the equilibrium thermal and wind field is conducted based on commercial CFD code ANSYS Fluent. The numerical settings include the SIMPLEC algorithm for pressure-velocity coupling, the central differencing scheme for diffusion terms, and the second-order upwind scheme for convective terms in the momentum equation and turbulence model equations.

The turbulence model employed is the SST $k-\omega$ model, where its transport equations for k and ω are derived by combining the inner layer Wilcox $k-\omega$ model multiplied by a hyperbolic tangent blending function F_1 and the outer layer standard $k-\epsilon$ model multiplied by $(1-F_1)$, as follows [33]:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i} \left[\rho u_i k - \left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_i} \right] = P_k - C_\mu \rho \omega k \quad (36)$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i} \left[\rho u_i \omega - \left(\mu + \frac{\mu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial x_i} \right] &= \gamma \frac{\omega}{k} P_k - \beta \rho \omega^2 \\ &+ 2(1-F_1) \frac{\rho}{\sigma_{\omega 2} \omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i} \end{aligned} \quad (37)$$

Where, P_k denotes the production of k , β and γ the constants in SST $k-\omega$ model, σ_k and σ_ω the Prandtl numbers of k and ω . Similarly, turbulence model constants in Eqs. (36) and (37) should be determined by F_1 , as follows:

$$\begin{aligned} \{\Phi\} &= \{\gamma, \beta, \sigma_k, \sigma_\omega\} = \{\Phi_1\} F_1 + \{\Phi_2\} (1-F_1) \\ \text{where } \{\Phi_1\} &= \{\gamma_1, \beta_1, \sigma_{k1}, \sigma_{\omega 1}\}; \{\Phi_2\} = \{\gamma_2, \beta_2, \sigma_{k2}, \sigma_{\omega 2}\} \end{aligned} \quad (38)$$

Where, $\{\Phi\}$, $\{\Phi_1\}$, $\{\Phi_2\}$ denote the constant sets of the SST $k-\omega$ model, the Wilcox $k-\omega$ model, and the standard $k-\epsilon$ model, respectively. According to the authors' previous studies [19,44], the turbulence model constants need to be modified appropriately to be more compatible with ABL characteristics rather than using default values. Here, as also suggested by the authors [20], $C_\mu = 0.028$; and $\gamma_1 = 0.413$, $\beta_1 = 0.0333$, $\sigma_{k1} = 1.176$, and $\sigma_{\omega 1} = 2$ for the inner layer Wilcox $k-\omega$ model, while $\gamma_2 = 0.413$, $\beta_2 = 0.0368$, $\sigma_{k2} = 1.0$, and $\sigma_{\omega 2} = 1.168$ for the outer layer

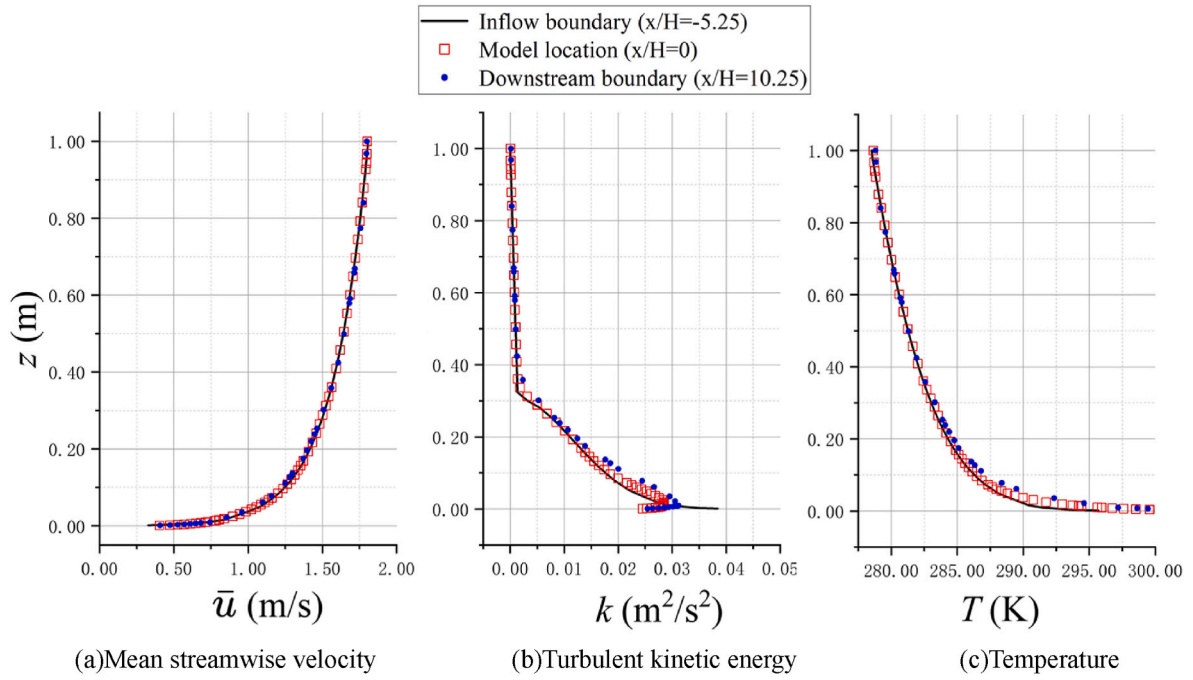


Fig. 4. Comparison of non-isothermal ABL profile at three different locations in case A (a)Mean streamwise velocity (b)Turbulent kinetic energy (c)Temperature.

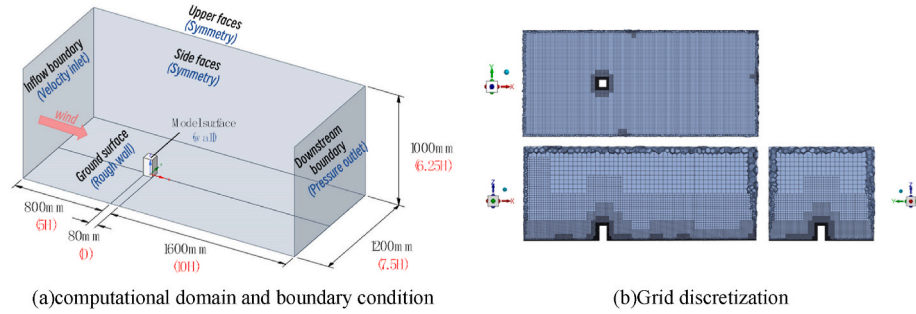


Fig. 5. Schematic diagram and information about Case B (a)computational domain and boundary condition (b)Grid discretization.

standard $k-\epsilon$ model.

The convergence criteria for steady numerical simulation are as follows: the scaled residuals for all variables should be below 10^{-12} ; while the scaled residual for the continuity equation should be below

10^{-8} ; and each scaled residual should show no further decrease during additional iterations. These convergence criteria are more stringent compared to previous studies, attributed to the better orthogonal quality of the grid.

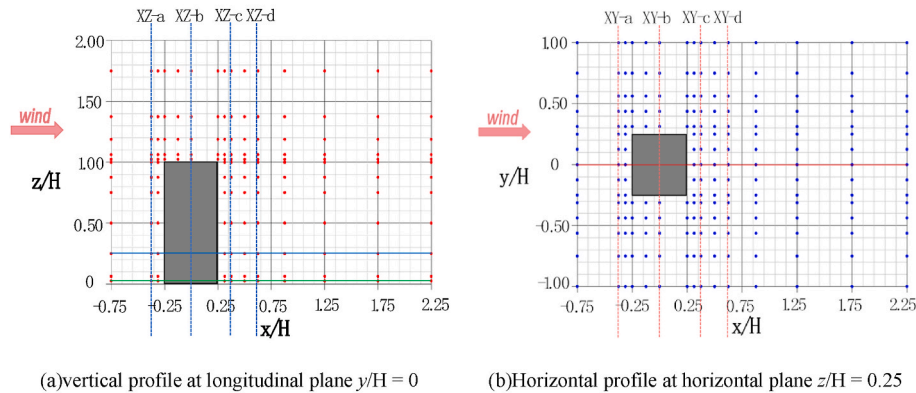


Fig. 6. Four locations ($x/H = -0.375, 0, 0.375, 0.625$) of wind tunnel test chosen for comparison (a)vertical profile at longitudinal plane $y/H = 0$ (b)Horizontal profile at horizontal plane $z/H = 0.25$.

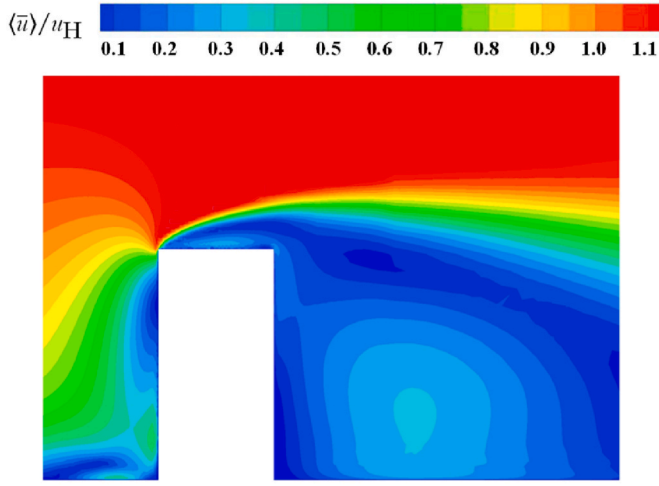


Fig. 7. Contour of $\langle \bar{u} \rangle / u_H$ at longitudinal plane ($y/H = 0$) of Case B (SST $k-\omega$ model).

3.4. Numerical results

Fig. 3 illustrate the numerical contour of the equilibrium thermal and wind field of Case A in longitudinal plane ($y/H = 0$). As can be seen in Fig. 3, the non-isothermal ABL flow reproducing by the equilibrium ABL model and the new equilibrium temperature inflow profile (Eqs. (31)–(35)) is sustained throughout the entire domain, achieving the horizontal homogeneity since the mean streamwise velocity $\bar{u}$, turbulent kinetic energy k , and temperature T show almost no streamwise gradient. As the flow progresses downstream, slight deviation in k and T occur in the small region near the ground surface, which is probably caused by the roughness model for the ground surface. In the commercial CFD code ANSYS Fluent [35], wall roughness is modeled by a simplified equivalent sand-grain model with K_s and C_s , which is essentially more applicable to pipe flow rather than the ABL. Even though the values of K_s and C_s are properly determined by referencing their relationship with z_0 derived by Blocken et al. [43], the simplified equivalent sand-grain model inevitably results in numerical errors near the ground surface.

Fig. 4 illustrate the vertical profiles of the equilibrium thermal and wind field extracted at the inflow boundary ($x/H = -5.25$), the hypothetical model location ($x/H = 0$), and the downstream boundary ($x/H = 10.25$). As can be seen in Fig. 4, the deviation of T caused by

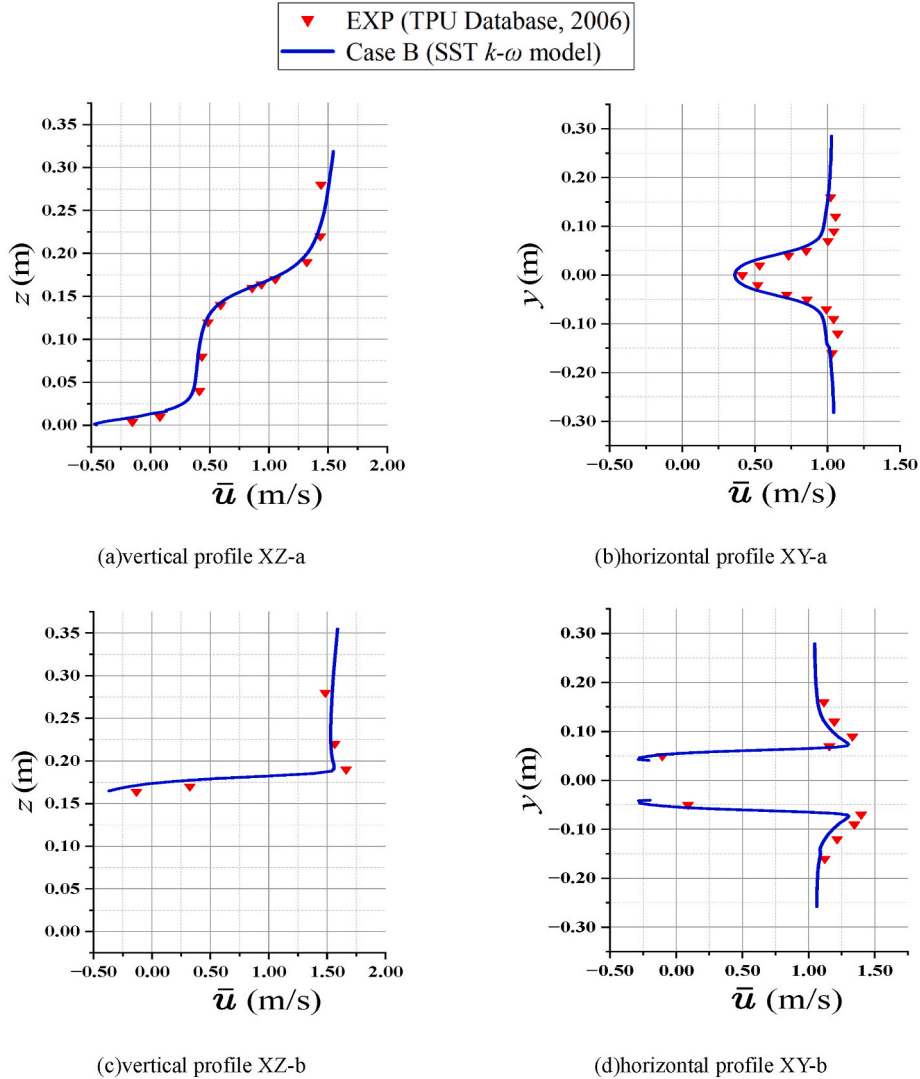


Fig. 8. Comparison of mean streamwise velocity profile (a)vertical profile XZ-a (b)horizontal profile XY-a (c)vertical profile XZ-b (d)horizontal profile XY-b (e)vertical profile XZ-c (f)horizontal profile XY-c (g)vertical profile XZ-d (h)horizontal profile XY-d.

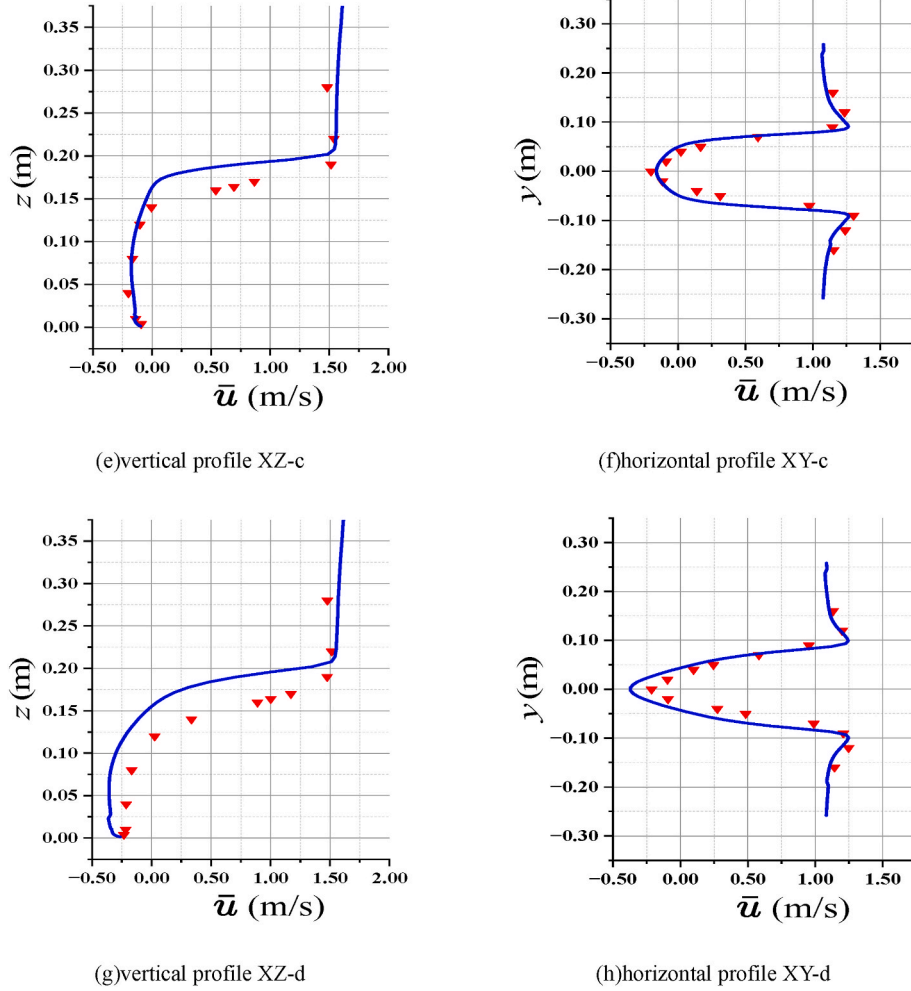
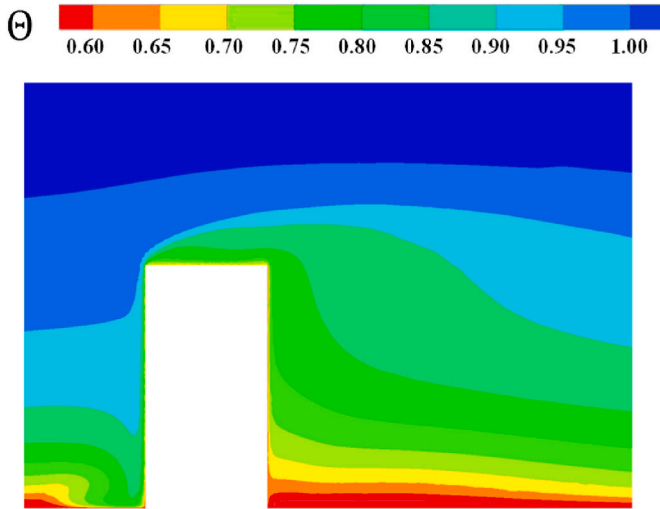


Fig. 8. (continued).

Fig. 9. Contour of normalized temperature on longitudinal plane ($y/H = 0$) of Case B (SST $k-\omega$ model).

horizontal inhomogeneity between the hypothetical model location and the inflow boundary does not exceed 0.5K, except for small regions near the ground surface ($z < 0.05$ m). In that region, as the flow progresses downstream, k decreases first and then increases, while T consistently

increases. Overall, the vertical profiles of $\bar{u}$, k , and T are sustained throughout all three locations, with almost no streamwise gradient. The numerical results verify the horizontal homogeneity of the non-isothermal ABL flow reproduced by the combination of the equilibrium ABL model and the new equilibrium temperature inflow profile shown in Eq. (19), laying the groundwork for further numerical simulations of the coupled UTWE.

4. Flow simulation application and result comparison

4.1. Numerical model and method

To validate the applicability of the present inflow profiles (Eqs. (31)–(35)) in simulating the coupled thermal and wind environment around buildings, a 3D numerical simulation was conducted in a computational domain containing a single square building model with dimensions of 0.16 m in height, 0.08 m in width, and 0.08 m in depth, denoted as Case B. As shown in Fig. 5(a), the computational domain for Case B is the same as for Case A, with the distance of $5H$ between the inflow boundary and the building model, and $10H$ between the building model and the downstream boundary, as suggested by the AIJ guide for CFD simulations [45]. Similarly, the boundary conditions listed in Table 1 were employed. Fig. 5(b) illustrates the 3D sectional view of grid discretization for Case B, whose grid discretization scheme is also the same as for Case A with a stretching ratio of 1.10, outlined in subsection 3.2. Additionally, the surface mesh size of the building model is set as 3 mm, ensuring a minimum of 25 cells on each edge. Overall, the total amount

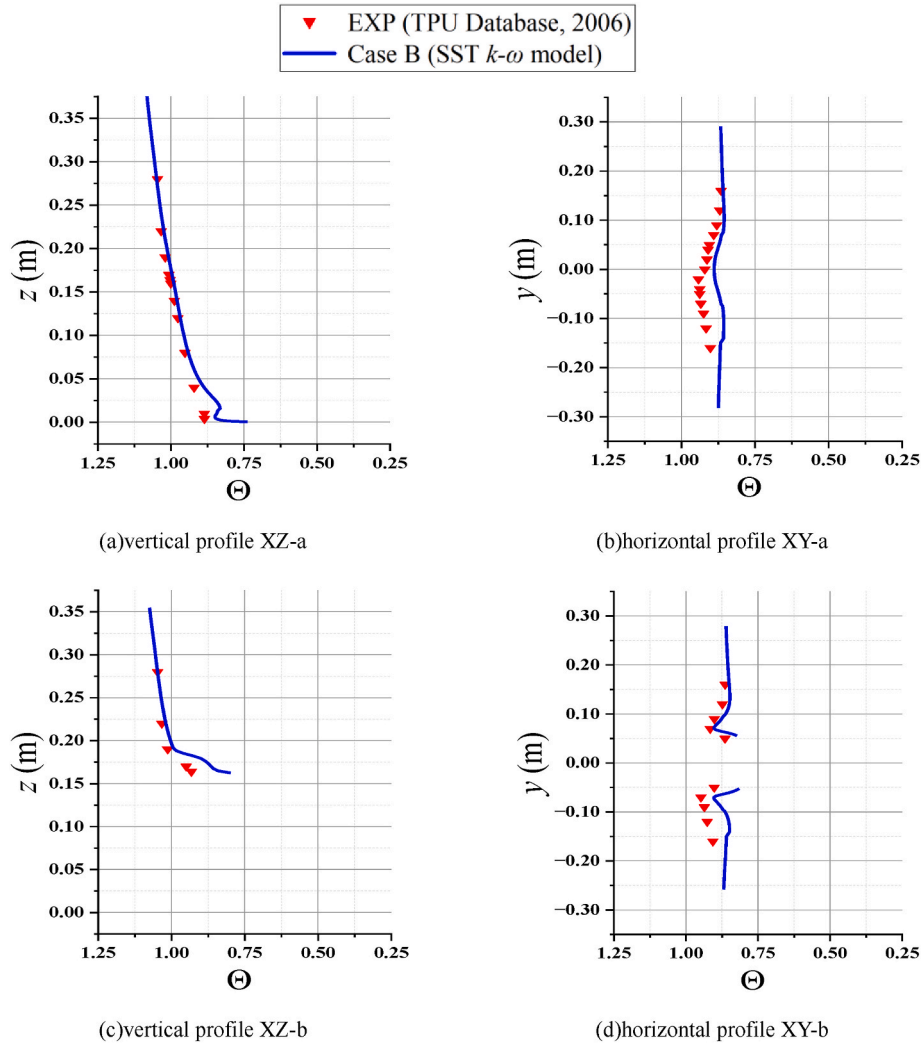


Fig. 10. Comparison of normalized temperature profile (a)vertical profile XZ-a (b)horizontal profile XY-a (c)vertical profile XZ-b (d) horizontal profile XY-b (e) vertical profile XZ-c (f) horizontal profile XY-c (g) vertical profile XZ-d (h) horizontal profile XY-d.

of cells of Case B is about 900,000, with all cells exhibiting orthogonality quality greater than 0.50. The numerical method and turbulence model constants depicted in subsection 3.3 were also employed to conduct steady numerical simulation.

4.2. Test information and locations chosen for comparison

In wind tunnel test conducted by TPU [42], the single square building model (0.16 m in height, 0.08 m in width, and 0.08 m in depth) was placed in the test section to study the thermal and wind environment around it under the non-isothermal ABL, with a corresponding blockage ratio of less than 3 %. As shown in Fig. 6, velocity and temperature were measured along the longitudinal plane at $y/H = 0$ and the horizontal plane at $z/H = 0.25$ during the test. Fig. 6 also illustrates four selected locations for extracting test results of the vertical and horizontal profiles of the UTWE for comparison, which are in front of and behind the building model and correspond to x/H values of -0.375 , 0 , 0.375 , and 0.625 , respectively.

4.3. Comparison of wind speed

Fig. 7 illustrates the numerical contour of the normalized mean streamwise velocity $\langle \bar{u} \rangle / u_H$ at the longitudinal plane ($y/H = 0$) in Case B,

in front of and behind the building model. As can be seen in Fig. 7, downdraft occurs in front of the building model due to its blockage effect, while updraft occurs in the recirculation region behind the building model due to the buoyancy effect.

Extracting numerical results of the vertical and horizontal profiles of mean streamwise velocity at locations shown in Fig. 6, compare them with the test results yielding Fig. 8. As can be seen in Fig. 8(a)–(b), the predicted mean streamwise velocity distribution of the downdraft matched the test results very well, where the absolute error is less than 0.05 m/s and the relative error is less than 5 % based on the reference wind speed of 1.37 m/s. This validates that the numerical simulation for Case B can accurately model the blockage effect of the building. As can be seen in Fig. 8(c)–(h), for the flow around the building model and the recirculation region, the mean streamwise velocity distribution can be predicted effectively. On one hand, the sharp vertical gradient of mean streamwise velocity produced above the roof ($z = 0.15$ – 0.20 m) was well predicted (as shown in Fig. 8(c)). While, it was underpredicted as the flow progresses in the recirculation region (as shown by Fig. 8(e) and (g)), resulting in significant deviations between the numerical and test results. But it is worth to note that the overall shape of their distribution consistently matched the test. On the other hand, as shown in Fig. 8(d)–(f), and (h), the predicted horizontal distribution of mean streamwise velocity in recirculation region matched the test results, where the

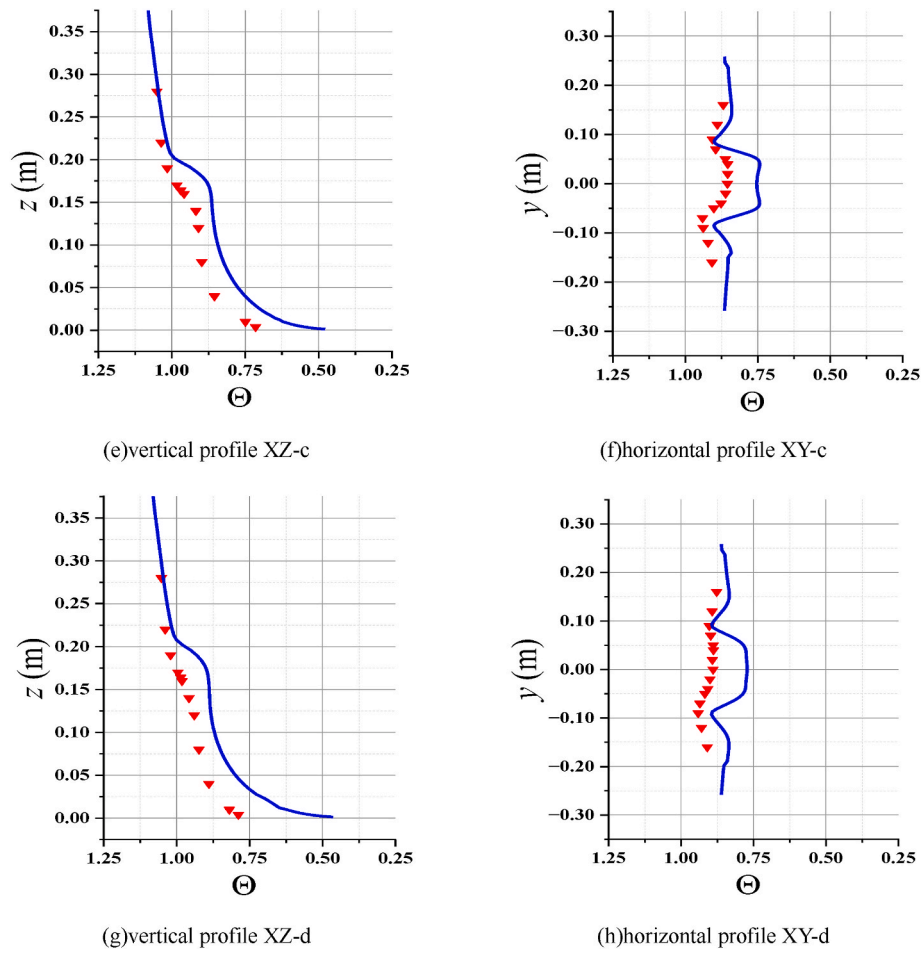


Fig. 10. (continued).

absolute error is less than 0.10 m/s and the relative error is less than 7.5 % based on the reference wind speed of 1.37 m/s.

4.4. Comparison of temperature

Fig. 9 illustrates the numerical contour of the normalized temperature Θ at the longitudinal plane ($y/H = 0$) in Case B, in front of and behind the building model. As can be seen in Fig. 9, the downdraft in front of the building model impacts the thermal stratification, leading to the reduction of the temperature and thus the increase of Θ . Then, the thermal stratification becomes complex in the recirculation region behind the building model, where heat accumulates around the lower corner near the ground, while the updraft consistently transports the heat upward, raising the temperature and thus decreasing Θ .

Similarly, extracting numerical results of the vertical and horizontal profiles of Θ at locations shown in Fig. 6, compare them with the test results yielding Fig. 10. As can be seen in Fig. 10(a)–(b), the predicted normalized temperature distribution of the downdraft matched the test results very well, where the absolute error is less than 0.1. This validates that the numerical simulation for Case B can accurately model the thermal stratification in front of the building. As can be seen in Fig. 10 (c)–(h), for the flow around the building model and the recirculation region, the Θ distribution can be basically predicted. Fig. 10(c)–(e), and (g) shows significant deviations between the numerical and test results behind the building model, indicating that Θ was overpredicted, i.e., the temperature was underpredicted. However, it is worth noting that the overall shape of the Θ distribution matched the test. In the recirculation region, as shown in Fig. 10(d)–(f), and (h), the comparison of numerical and test results reveals that the Θ deviation gradually decreases from

longitudinal location towards both sides. Overall, the absolute error of Θ in Case B is within the range of 0.10–0.15.

5. Conclusions

Based on the equilibrium ABL model, this study theoretically derives a new equilibrium temperature inflow profile compatible with the energy equation, the standard gradient diffusion hypothesis, and the turbulence model, while ensuring horizontal homogeneity. The major conclusions are as follows:

- (1) The new equilibrium temperature inflow profile theoretically coincides with the actual temperature distribution of the non-isothermal ABL flow. It is proportional to the turbulent kinetic energy inflow profile, with the ratio further determined by wall heat flux, demonstrating that turbulent thermal diffusion is strongly related to turbulent characteristics and atmospheric stability. To address the scaled effect, its full expression is derived based on similarity criteria of the normalized temperature and the bulk Richardson number.
- (2) Using a numerical model of empty domain based on the SST $k-\omega$ model, the horizontal homogeneity of the equilibrium thermal and wind field modeled by the new set of inflow profiles is verified. The profiles of $\bar{u}$, k , and T are sustained throughout the entire domain with almost no streamwise gradient, laying the groundwork for further numerical simulations of the coupled UTWE.
- (3) Taking the 3D numerical simulation for the flow around a single square building model as an example, the applicability of the new

equilibrium temperature inflow profile is validated. The comparison between numerical and test results illustrates that, the thermal and wind environment around the building can be effectively predicted, where the relative error of mean stream-wise velocity is less than 7.5 % and the absolute error of normalized temperature is within the range of 0.10–0.15.

CRediT authorship contribution statement

Daoqi Chen: Writing – original draft, Validation, Formal analysis, Data curation. **Yi Yang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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